

Algorithmic Pseudocode Sample 26

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Require: Sequence reads $X_i = \{X_0, X_1, \dots, X_N\}, \forall X_i \in \mathcal{X}_i$

Ensure: Pseudo-image I_r

procedure RELATIVECOOCCURRENCE(x_i, x_j, I_r)

$e_{i,j} \leftarrow I_r[i, j]$

 ▷ Current co-occurrence frequency

$e'_{i,j} \leftarrow e_{i,j} + 1$

 ▷ New co-occurrence frequency

$r \leftarrow \frac{e_{i,j}}{(\|e_{i,j} - e'_{i,j}\|_2)}$

 ▷ Update co-occurrence

$e_{i,j} = 2\sqrt{\max(r - 1, 1)} + (\min(r - 2, 2) + 2)$

return $e_{i,j}$

 ▷ Relative co-occurrence frequency

end procedure

procedure NORMALIZEIMAGE(I_r, λ_m)

$N \leftarrow \sum_{i=0}^{4^k} \sum_{j=0}^{4^k} I[i, j]$

for $i \in 4^k$ **do**

for $j \in 4^k$ **do**

if $I_r[i, j] > \lambda_m$ **then**

$I_r[i, j] \leftarrow 255 * I[i, j] / N$

else

$I_r[i, j] \leftarrow 0$

end if

end for

end for

return I_r

 ▷ Normalized Pseudo-Image

end procedure

$I_r \leftarrow \text{ones}(4^k, 4^k)$

 ▷ Initial pseudo-image

for $X_i \in \mathcal{X}_i$ **do**

 ▷ Iterate through each sequence

for $x_i, x_j \in X_i$ **do**

 ▷ Iterate through successive k-mers

$I_r[i, j] \leftarrow \text{RELATIVECOOCCURRENCE}(x_i, x_j, I_r)$

end for

end for

$I_r \leftarrow \text{NORMALIZEIMAGE}(I_r, \lambda_m)$

return I_r

 ▷ Final pseudo-image
